F2: tan(x)

Implemented from First Principles

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The Problem

Compute $\tan(x)$ without any library performing the mathematics. No `math` module, no built-in factorial, no built-in trigonometry.

The constraint is what makes the work interesting: every decision a library would have made silently must now be made explicitly, and each becomes a requirement that can be verified.

The Algorithm

Maclaurin series for both components of the quotient identity:

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

No factorial is ever computed. Dividing one term by its predecessor gives the recurrence

$$t_{k+1} = t_k \cdot \frac{-x^2}{(2k)(2k+1)}$$

which costs one multiplication and one division per term and cannot overflow.

Range reduction. The tangent has period π, not 2π, because **sin** and **cos** both change sign and the signs cancel in the ratio. Subtracting the nearest whole multiple of π confines the argument to $[-\pi/2,\ \pi/2]$, where the series converges quickly and every asymptote collapses onto a single check.

Termination by accuracy, not by count. Summation stops when a term falls below 10^{-15} — the point at which it no longer changes the result at double precision.

A Defect Found by Measurement

Testing against a reference at $x = 10^{15}$ returned -1.557 where the true value is -1.672 — wrong in the first decimal, with no error raised.

Sampling four thousand angles at each order of magnitude showed roughly one significant digit lost per factor of ten:

Magnitude	Typical	Worst
10^2	13.6	10.8
10^4	11.7	8.8
10^6	9.7	6.8
10^8	7.7	5.1

Results display ten decimal places. Past 10^6 the worst case falls below six correct digits, so the display would present noise as precision. **The validated range is $\pm 10^6$**; larger angles are refused with an explanation rather than answered.

All 636,620 exact asymptotes within that range were tested. None was missed.

Unit Testing (PyUnit)

58 tests across eleven classes, each traced to a requirement from Deliverable 2. Six of the eleven:

Class	Requirement
Reference angles	TAN-FR-01, 03
Range reduction	TAN-FR-04
Undefined points	TAN-FR-07
Validated range	TAN-FR-11
Termination	TAN-FR-06
From-scratch constraint	TAN-FR-02

Validated by mutation. Sixteen defects were injected. The first version of the suite detected **seven**; the revised version detects **fourteen**. Both survivors are behaviour-preserving.

The nine original survivors shared one cause: every tuning constant was asserted only relative to itself. Loosening the series tolerance degraded accuracy by four orders of magnitude and nothing failed. The constants are now pinned to their documented values.

Quality Tooling

```
(base) brandon@Konans-MacBook-Pro source_code % cd ~/Downloads/soen6011-tan-calculator/source_code

flake8 --max-line-length=79 tan_math.py tan_gui.py verify_tan.py && echo "Flake8: 0 issues"
pylint tan_math.py
pylint tan_gui.py
pylint verify_tan.py
Flake8: 0 issues

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)


--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)


--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)

(base) brandon@Konans-MacBook-Pro source_code %
```

Flake8, 79-column limit	no issues
Pylint, all three modules	10.00 / 10
Unit tests	58 of 58
Verification harness	44 of 44

Debugger: Terms Shrinking to Tolerance

```
(base) brandon@Konans-MacBook-Pro source_code % python -m pdb pdb_demo.py
> /Users/brandon/Desktop/soen6011-tan-calculator/source_code/pdb_demo.py(1)<module>()
-> """Driver for the pdb demonstration."""
(Pdb) break tan_math.py:242
Breakpoint 1 at /Users/brandon/Desktop/soen6011-tan-calculator/source_code/tan_math.py:242
(Pdb) continue
> /Users/brandon/Desktop/soen6011-tan-calculator/source_code/tan_math.py(242)sine()
-> term *= -x * x / ((2.0 * k) * (2.0 * k + 1.0))
(Pdb) p k, term, total
(1, 0.9735361584457678, 0.9735361584457678)
(Pdb) continue
> /Users/brandon/Desktop/soen6011-tan-calculator/source_code/tan_math.py(242)sine()
-> term *= -x * x / ((2.0 * k) * (2.0 * k + 1.0))
(Pdb) p k, term, total
(2, -0.1537818244191063, 0.8197543340266615)
(Pdb) continue
> /Users/brandon/Desktop/soen6011-tan-calculator/source_code/tan_math.py(242)sine()
-> term *= -x * x / ((2.0 * k) * (2.0 * k + 1.0))
(Pdb) p k, term, total
(3, 0.007287510376427245, 0.8270418444030888)
(Pdb) continue
> /Users/brandon/Desktop/soen6011-tan-calculator/source_code/tan_math.py(242)sine()
-> term *= -x * x / ((2.0 * k) * (2.0 * k + 1.0))
(Pdb) p k, term, total
(4, -0.0001644500722499108, 0.826877394330839)
(Pdb) p abs(term) < TOLERANCE
False
(Pdb) quit
(base) brandon@Konans-MacBook-Pro source_code %
```

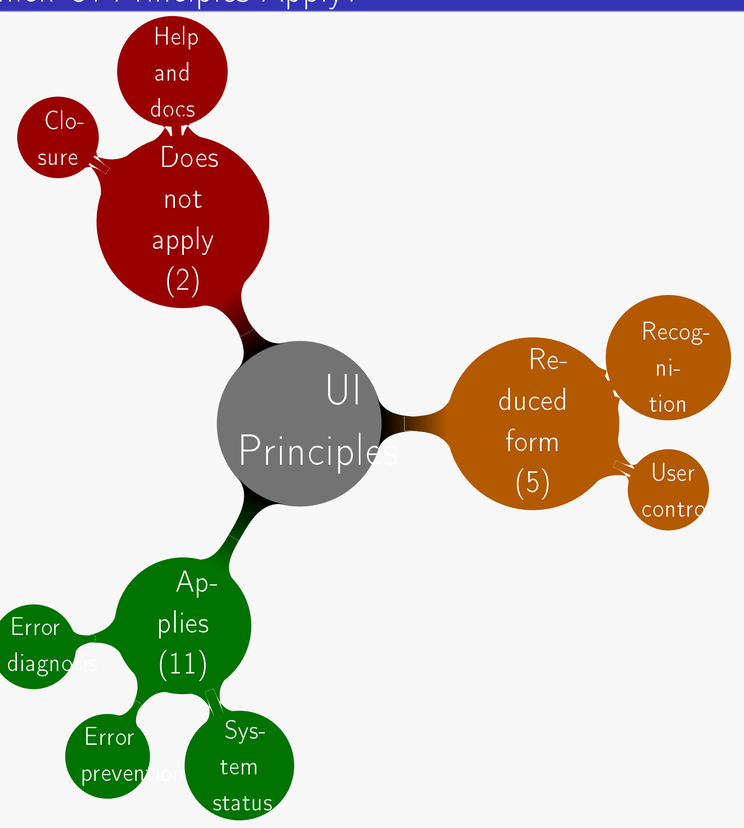
Successive terms fall $0.97 \to 0.15 \to 0.007 \to 0.0001$. The stopping rule is visible rather than asserted.

What the Faults Had in Common

Four defects surfaced: silent precision loss at large arguments, a signature mismatch between the modules, a documented command that failed on a fresh clone, and an unreachable validation branch.

Every one appeared only when something was run. None was visible by reading.

Which UI Principles Apply?



Nielsen's heuristics and Shneiderman's rules were assessed individually. Two do not apply: a **help system** for six labelled controls would be evidence the labels failed, and **closure** presupposes a sequence an atomic task lacks.

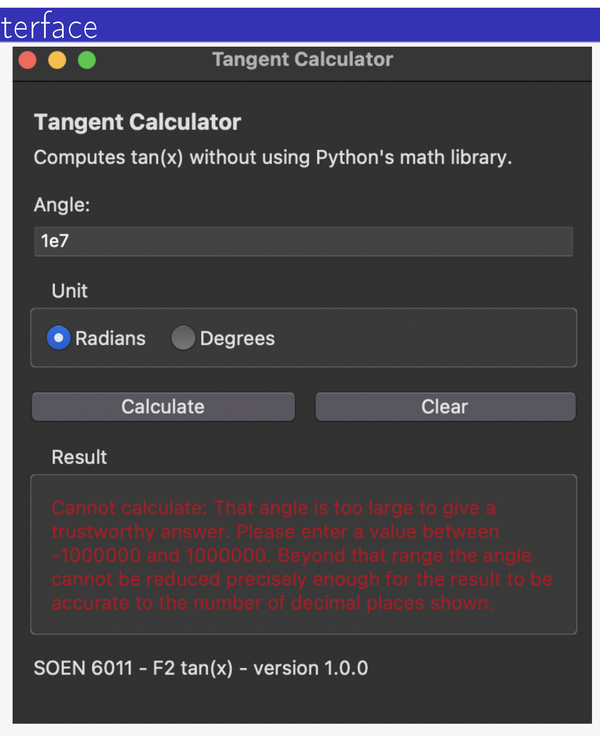
Accessibility

The interface aims to be accessible at the level the framework permits:

- Every control carries a visible text label
- All functionality reachable from the keyboard; Return calculates, Escape clears
- Focus begins in the entry field and is visibly indicated
- **Every outcome is stated in words as well as colour**, so nothing depends on colour perception
- **Contrast measured on both themes.** No single colour can meet 4.5:1 against a light and a dark background — the required luminance ranges do not overlap — so the palette is selected at run time from the theme background. Ratios are 5.8–7.1:1 on light and 5.7–7.0:1 on dark
- Text wraps rather than clipping; system font settings apply

Stated limitation. Screen-reader conformance is not claimed: Tkinter exposes no accessibility tree on any platform, so WCAG 4.1.2 and 4.1.3 cannot be met within this framework. That is a property of Tkinter, and is recorded rather than worked around.

The Interface



An angle beyond the validated range, refused with an explanation rather than a fault code.

Generative AI Use

Four CASTROFF-structured prompts informed this deliverable: a chain-of-thought analysis of which UI principles apply, an accessibility audit against WCAG 2.1, a test-suite review, and a debugger session design.

Two reviews changed the code. The accessibility audit measured the colours against a dark theme and found all three below 2:1; the palette is now chosen at run time. The test review found nine mutations surviving; the suite now catches fourteen of sixteen. Prompts and responses are in `screenshots/`.

Repository

`github.com/Brand0-code/`

`soen6011-tan-calculator`

Version `v1.0.0` — Semantic Versioning